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Two distinct patterns of acne progression: Pathogenesis and implications for management

Acne vulgaris, the 8th most common global disease,[1] has an incidence of 64.95 per 100,000 (Global Burden of Disease 2021).[2] Previous studies lacked dynamic studies on exacerbations of acne progression.[3] In clinical practice, we have observed that patients with acute exacerbations or new onset (elevated severity) and those with long-term cyclical activity (similar severity levels) display distinct triggers and clinical presentations, even when presenting with the same severity grade at consultation.

To explore the characteristic differences between the 2 patterns, we conducted a comprehensive questionnaire survey after obtaining approval from the institutional review board, involving 350 participants, achieving a 91.7% response rate (321/350). Acne vulgaris was diagnosed and categorized into grades I to IV based on clinical presentation, with differential diagnoses, such as acneiform drug reactions, chloracne, rosacea, and perioral dermatitis, etc. Based on their recollections of acne flareups over the past year, patients were divided into 2 groups: the breakthrough group (183 patients) experienced a sudden outbreak or first-time occurrence with a >30% increase in lesion count, significantly elevating their severity level from the baseline within 3 months, and the periodic group (138 patients), with monthly relapses and <10% increase in lesion count. (Fig 1)

When compared with the periodic group, the breakthrough group is characterized by exacerbated involvement of the perifacial zones (especially the cheeks), excluding the nose. This contradicts the common belief that the nose, due to its high-sebum production, is a primary site for acne flareups. In contrast, the periodic group shows minimal change in lesion frequency across facial zones from baseline. (Fig 1)

Multivariate logistic regression identified acute endogenous stress behaviors that routine habit changes (eg, moving/traveling/business trips, odds ratio [OR] = 1.998, $P = .009$), increased psychological stress (OR = 2.075, $P = .004$), frequent late-night staying up (OR = 1.914, $P = .010$), and high-fat diet consumption (OR = 2.038, $P = .004$) as independent risk factors for the breakthrough onset pattern. (Supplementary Tables I and II, available via Mendeley at https://data.mendeley.com/datasets/3z9zvphkv8/1).

Previous literature have shown that endogenous factors suggest the involvement of oxidative stress in the pathogenesis of acne.[4] Therefore, we validated the clinical classification by measuring serum oxidative (total oxidant status, protein carbonyl, and malondialdehyde) and antioxidative (total antioxidants content, catalase, glutathione peroxidase, and superoxide dismutase) markers in 128 patients across 2 groups. The breakthrough group exhibit significantly higher levels of oxidative stress than the periodic group. (Supplementary Figs 1 and 2, available via Mendeley at https://data.mendeley.com/datasets/3z9zvphkv8/1).

Our study identified 2 acne progression patterns with distinct pathogenic features and management implications. Acne inflammation induced by acute endogenous stress is characterized by the following: (1) rapid response to triggers, (2) perifacial lesion predominance, and (3) may not parallel sebum secretion. These findings suggest potential involvement of unknown pathogenic factors. For the breakthrough group, short-term potent antioxidants are crucial. For the periodic group, early isotretinoin use may be more appropriate, which provides a more comprehensive coverage of the pathogenesis of acne. The recently adopted strategy of using genetically modified Cutibacterium[5] to modulate cutaneous sebum production may be more important for the periodic group. However, therapies targeting unknown mechanisms mentioned above may offer greater clinical value. (Table I)

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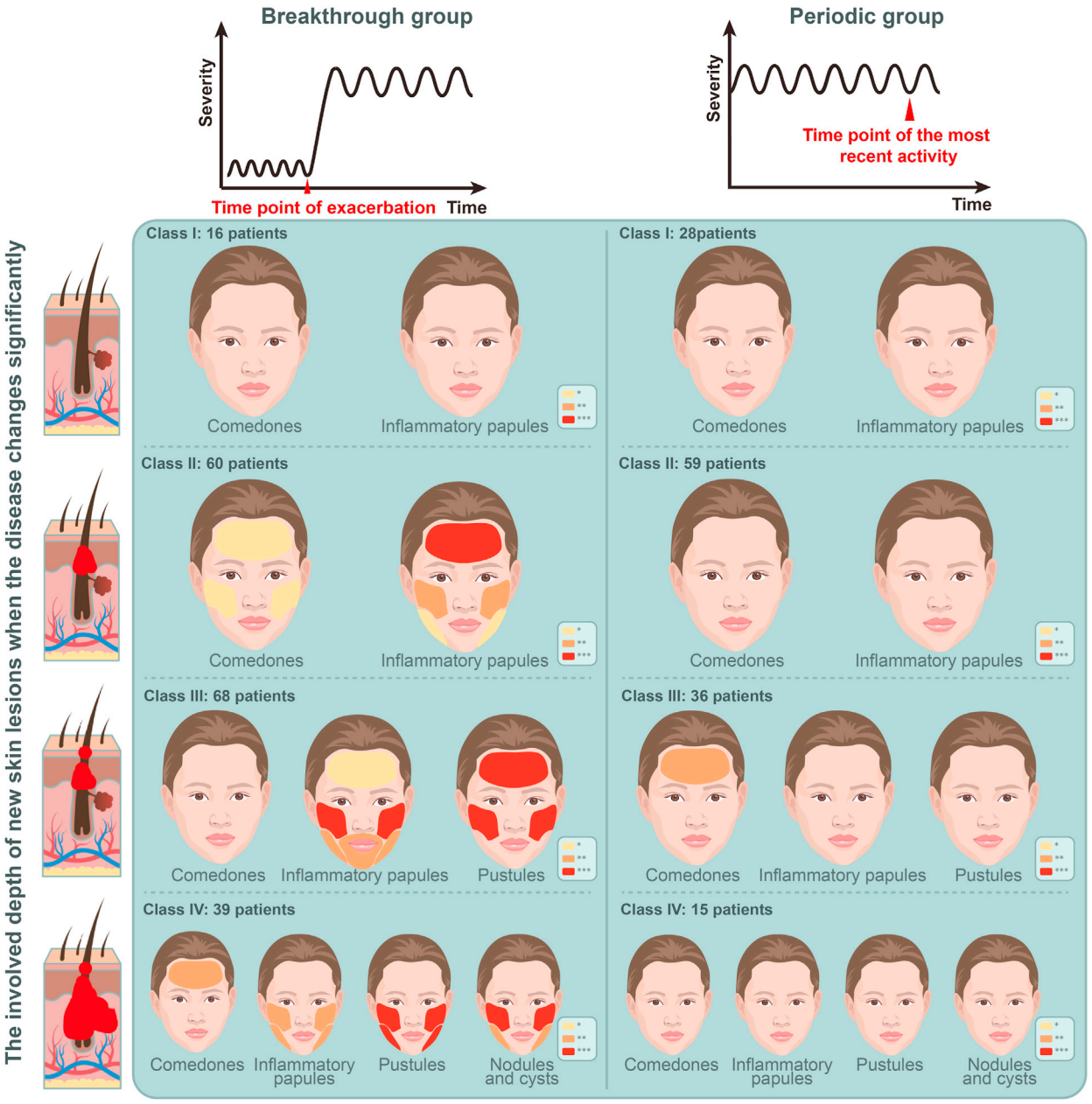


Fig 1. Acne vulgaris condition patterns in the breakthrough group and the periodic group. Differences in the frequency of various skin lesions in different facial zones before and after condition changes. (*$P < .05$, **$P < .01$, ***$P < .001$).

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Table I. Identification and management of acne vulgaris in the breakthrough and periodic groups

	Characteristic		Oral drug treatment			
Patterns of acne	**Manifestations**	**Triggers**		**Grade II**	**Grade III**	**Grade IV**
			First-line	Antibiotics + antioxidants	Antibiotics + antioxidants	Antibiotics + antioxidants
Acute exacerbations or new onset (breakthrough group)	Perifacial zones predominantly (especially the cheeks)	Change routine habits, such as moving/traveling/business trips, increased psychological stress, more frequent staying up late, and consuming a high-fat diet	Second-line	Antibiotics + antioxidants	Antibiotics + antioxidants + isotretinoin	Antibiotics + antioxidants + Isotretinoin
Similar severity levels (periodic group)	Cheeks may not be predominantly involved (the perioral or T-zone may predominantly)	Menstrual cycle, etc.		Isotretinoin + antiandrogenic drugs (indicated for women) + genetically modified Cutibacterium	Antibiotics + antioxidants + isotretinoin + antiandrogenic drugs (indicated for women) + genetically modified Cutibacterium	Antibiotics + antioxidants + isotretinoin + antiandrogenic drugs (indicated for women)

IRB approval status: Reviewed and approved by Chengdu Medical College (approval #2021CY-FYIRB-BA-42-01).

Key words: acne vulgaris; clinical features; exacerbation; facial zone; inflammation; oxidative stress; onset patterns; risk factors.

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Conflicts of interest

None disclosed.

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Isotretinoin laboratory monitoring varies by sex for patients with acne vulgaris

Isotretinoin is a highly effective treatment for severe and recalcitrant acne vulgaris.[1] A recent Delphi consensus study recommended checking alanine aminotransferase and triglycerides within 1 month prior to initiation and at peak dose owing to concern for associated transaminitis and hypertriglyceridemia.[2,3] Isotretinoin is teratogenic, and monthly pregnancy testing is required for individuals who can become pregnant; yet, data on types of pregnancy screening have not been reported. This study characterizes current trends in pregnancy testing for patients receiving isotretinoin and differences in laboratory monitoring by sex.

Population-level data were extracted from TriNetX, representing 212 million patients in the United States. The study period was from January 2017 until December 2021 for patients aged 12 to 30 years receiving isotretinoin for a diagnosis of acne vulgaris. Statistical analyses were performed using Student *t* tests for quantitative variables and χ^2 for categorical variables.

In total, 809,999 isotretinoin fills were generated for 165,745 unique patients (52% male and 48% female). The mean age at first fill for isotretinoin was 18.4 years with a mean duration of treatment of 4.9 months. For female patients, the mean number of pregnancy tests was 4.43. For in-office pregnancy tests, 90.3% were with urine, whereas 75.4% of laboratory pregnancy tests were with serum (Table I). The mean numbers of complete blood count, lipid, and liver function tests were greater for female patients than for male patients ($P < .001$) (Table II).

This study shows that a substantial number of providers order serum pregnancy tests for their patients. Although a baseline serum pregnancy test would be reasonable if other laboratory studies are being drawn, monthly blood draws for pregnancy testing result in greater risk and cost to the patient. Interestingly, the greatest increase in use of in-office urine pregnancy testing occurred between 2020 and 2021, which coincides with the COVID-19 pandemic when iPLEDGE pregnancy testing restrictions temporarily were eased. The Clinical Laboratory Improvement Amendments' requirement for pregnancy testing was permanently revoked in November 2023.[4] Subsequently, the percentage of laboratory pregnancy tests decreased over the study period, whereas office pregnancy tests increased.

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